

ANALYZING QUESTIONNAIRE DATA

Spring 2026

Instructor: Corinna Noel, PhD

What do you remember about best practices for questionnaire design?

- **Begin with clear objectives:** Questions must directly reflect research questions/hypotheses
- **Limit to essential data:** Collect only variables needed for analysis
- **Use or adapt validated questionnaires** when possible
- **Keep questions clear & specific:** Simple language; avoid jargon, leading, or double-barreled items
- For quantitative inquiry, favor closed-ended formats: Ensure response options are **mutually exclusive and exhaustive**
- **Order for flow & comfort:** General → specific; easy → hard; non-threatening → sensitive
- **Design with analysis in mind:** Use consistent formats; collect continuous data when feasible
- **Test, train, and refine:** Pilot test; train interviewers (if appropriate) to ensure reliability and validity

Why is it important to know how to design and analyze questionnaire data as a public health professional?

Overview of class

- Common types of questions on questionnaires + why this matters for analysis
- Review of types of variables
- Setting the stage for the example we will be working with today

- *BREAK* -

- Implementation tips for online questionnaires
- Developing analysis plans
- Data dictionaries
- Cleaning data + inclusion/exclusion criteria

- *BREAK* -

- Example statistics using questionnaire data: Descriptive, bivariate, + multivariable models with control for confounding

... with activity/discussion prompts + R exercises implemented throughout

Common types of questions on questionnaires

1. Closed-Ended Questions

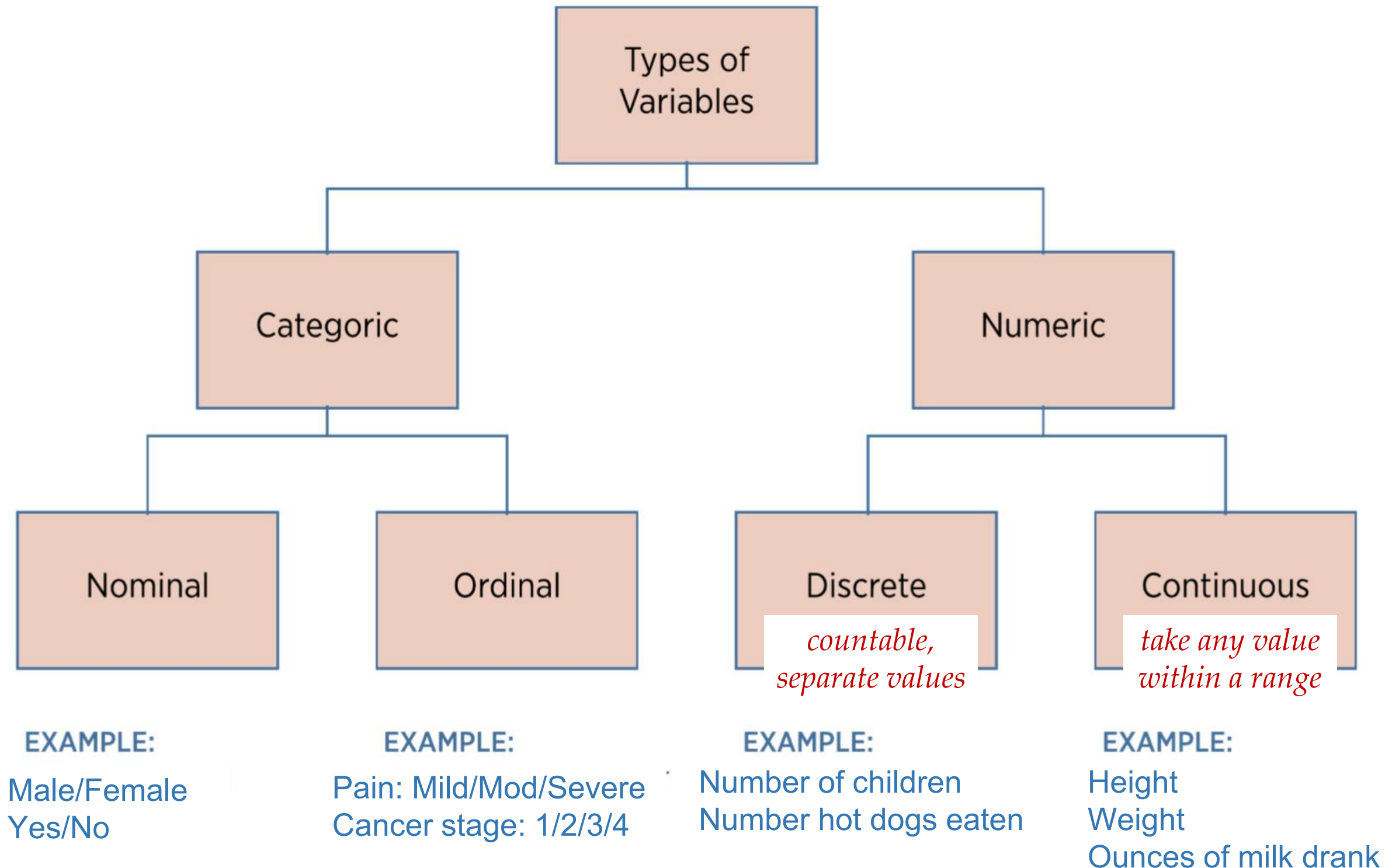
- Respondents choose from a fixed set of response options that are mutually exclusive and exhaustive.
- Common formats include:
 - Yes/No (binary)
 - Multiple choice (single response)
 - Check-all-that-apply (multiple response)
 - Rating-scale questions (e.g., Likert, frequency scales)
 - Ranking questions (ordering items in a list by preference or importance)
- Strengths: Easy to answer, code, and analyze; higher reliability
- Limitations: May miss nuance or unanticipated responses

2. Open-Ended Questions → *will not be covered in this lecture*

- Respondents answer in their own words; Free-text responses
- Strengths: Useful for exploratory inquiry and capturing unanticipated responses
- Limitations: Greater respondent burden; more difficult to code and analyze

Remember: How data is recorded

... affects types of analyses that can be performed!



Important to note

- Some variables can be analyzed on a categoric or numeric scale, depending on data + analytic goals
- For example:
 - Ordinal variable: Likert scale assessing the effectiveness of dashboard in disseminating data and resources
 - Numeric variable
 - Discrete: Number of COVID-19 hospital admissions (*whole numbers*)
 - Continuous: Weight (in pounds; *precise to 1 decimal point*)

Consider these Likert scales

Overall, how effective do you think the Cayuga County Overdose Data Dashboard is in...?

	Very effective	Effective	Fairly effective	Slightly effective	Not effective
1) Sharing data on overdoses in the county	5	4	3	2	1
2) Sharing information on resources related to substance use in the county	5	4	3	2	1

	start_date	progress	duration_in_seconds	effective_data
1	2023-04-27 19:16:00	100	387	Very effective
2	2023-04-28 09:46:00	100	224	Very effective
3	2023-05-01 12:48:00	100	155	Very effective
4	2023-05-02 13:33:00	100	263	Fairly effective
5	2023-05-03 08:06:00	100	273	Effective
6	2023-05-03 08:18:00	100	527	Very effective
7	2023-05-03 09:11:00	100	429	Effective
8	2023-05-03 09:09:00	100	535	Very effective
9	2023-05-03 09:09:00	100	820	Effective
10	2023-05-03 09:37:00	100	212	Fairly effective
11	2023-05-03 10:43:00	100	257	Very effective
12	2023-05-03 09:44:00	100	5514	Effective

Decision on categoric or numeric in analysis depends on:

(1) Analytic goals:

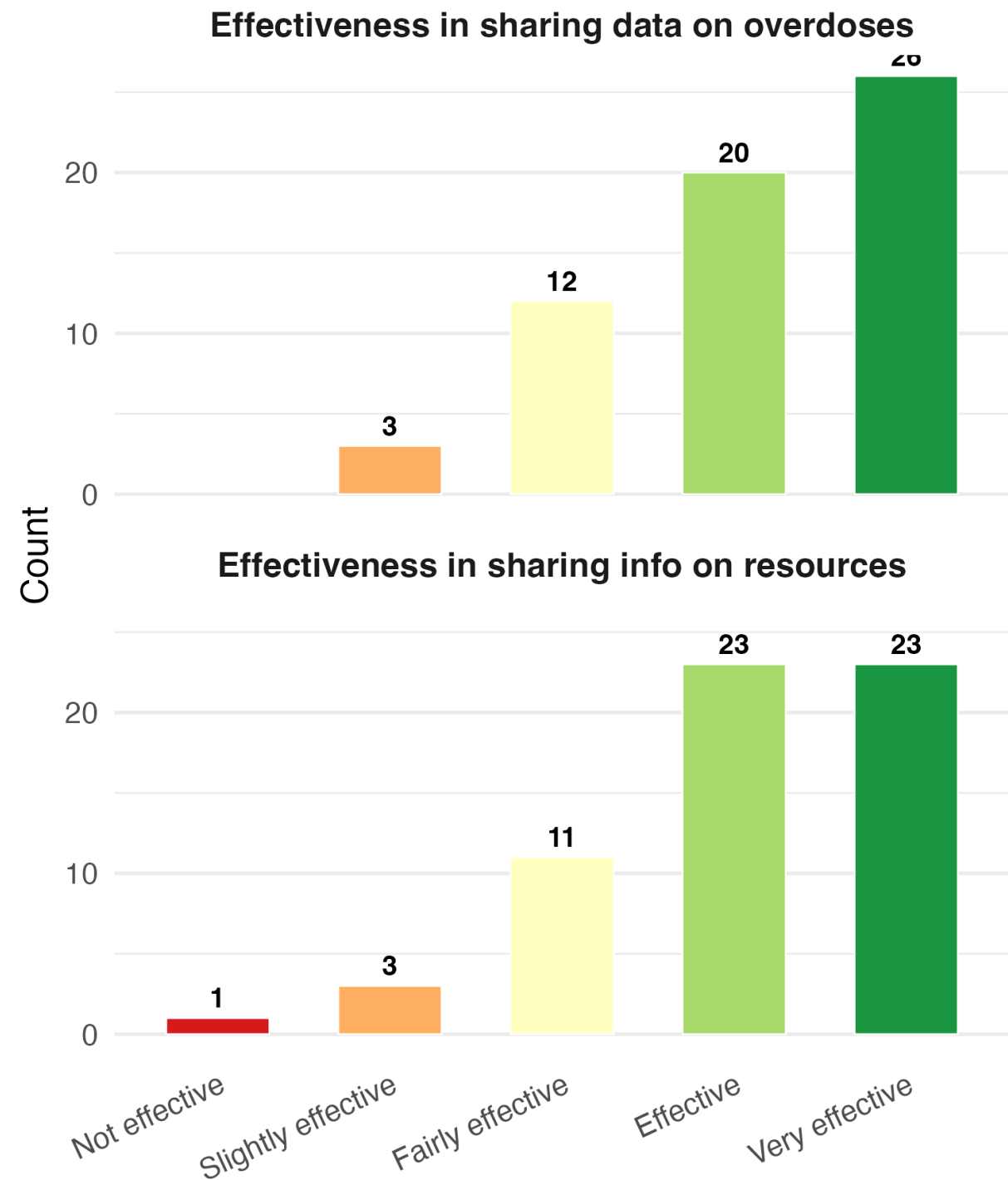
- Treat as **categorical** when goal is to describe response patterns → "79% ($n=48$) rated the dashboard as 'Effective' or 'Very effective' in sharing data on overdoses; 75% ($n=48$) for sharing information on resources"
- Treat as **numeric** when goal is to summarize or compare → "The mean effectiveness rating was 4.3 ($SD = 0.8$) out of 5 for sharing data on overdoses"

(2) Distribution of data:

- **Always visualize first** → the shape of your data should drive your analytic choice
- Both distributions here are left-skewed, with most responses in *Effective* and *Very effective* → favors categorical reporting (n and %)
- A mean alone is misleading when data are skewed
- If responses were spread more evenly across all five categories, numeric treatment and mean comparisons would be more appropriate and informative

For these data, recommended to:

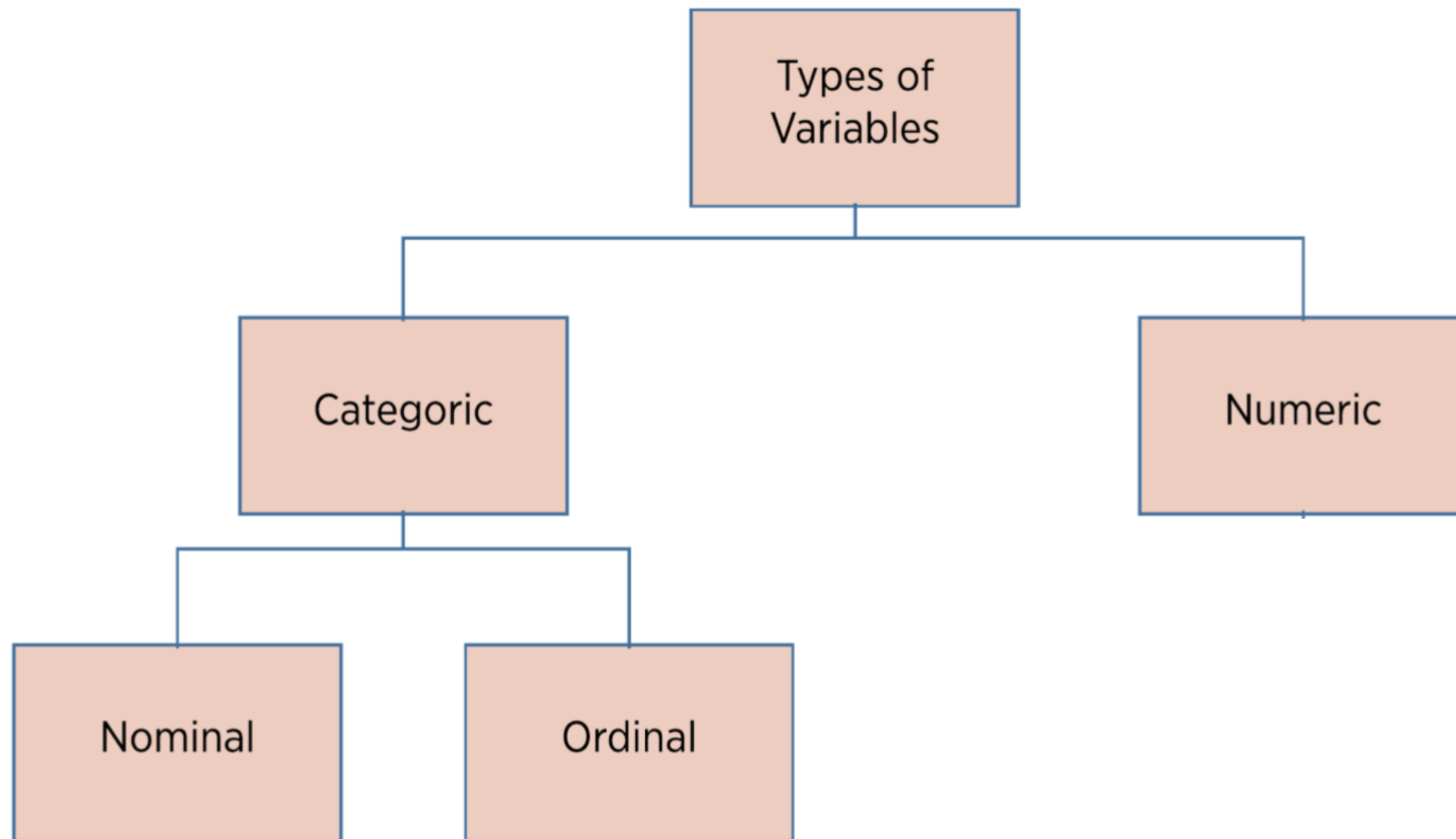
- Report categorically first: "The majority of respondents (79%, $n=48$) rated the dashboard as effective or very effective in sharing overdose data", then add the mean and SD as a summary statistic



Important to note

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 - Numeric variable
 - **Discrete: Number of COVID-19 hospital admissions** (*whole numbers*)
 - **Continuous: Weight** (in pounds; *precise to 1 decimal point*)

Main point



Be aware of the type of data you will collect as you develop questions in a questionnaire, as this will impact *how you can analyze your data*

Today's topic: Wildfire smoke in NYS in summer 2023

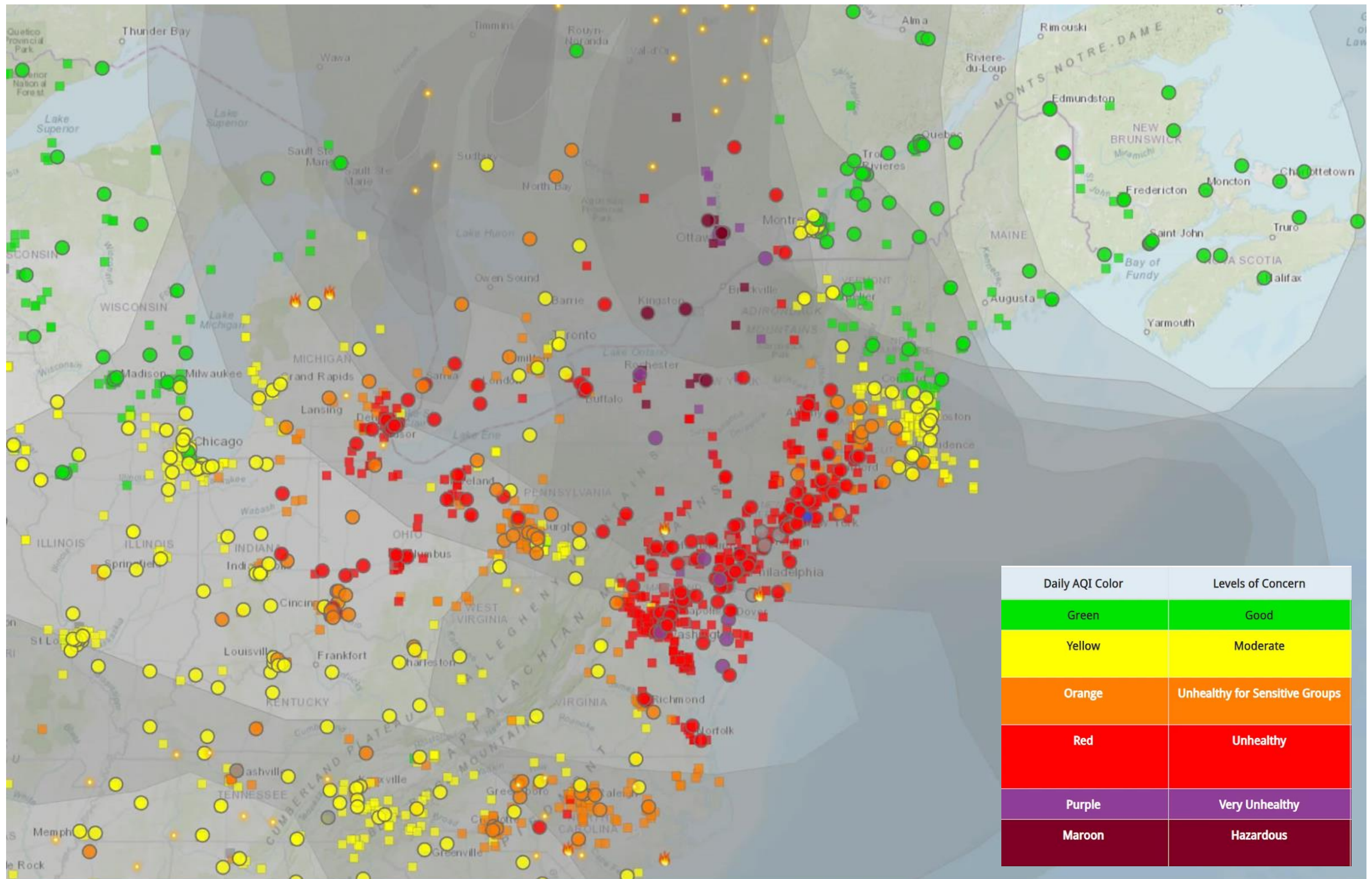


Why this topic is important

- **Wildfire smoke (WFS) is a growing public health issue** driven by climate change, drought, extreme weather, fire suppression practices, and expansion of the wildland–urban interface, increasing population exposure globally and in the U.S. (1–4)
- **WFS contains elevated PM_{2.5}**, which penetrates deep into the lungs and may be more harmful than PM_{2.5} from other sources, contributing to respiratory, cardiovascular, and mental health impacts, particularly among vulnerable populations (5–15)
- **Despite expanding wildfire impacts, most U.S. WFS research has focused on western states**, with limited evidence on other regions (16–21)
- **The 2023 Canadian wildfires resulted in unprecedented, sustained WFS exposure in the northeastern U.S., including New York State**, creating a novel context to assess perceptions, health impacts, and protective behaviors, building on prior research (22–25)

See reference list on last slide

Wildfire smoke from Canada: Plumes on June 7, 2023



fire.airnow.gov

Air quality in NYC on June 7 vs. June 9, 2023



[Twitter / X](#)

Our study



Public Health



Did you live in the Northeast US during the summer 2023 wildfire season?

Public health researchers at Cornell University want to hear about your experience with wildfire smoke!

What will you be asked to do?

Take a 10-minute survey.

Upon completion, you will have the chance to enter a raffle for one of five **\$50.00 Amazon gift cards**.

Who is eligible?

Anyone aged 18+ years, who spent time in the Northeastern US during the summer 2023 wildfire season

bit.ly/wildfire_smoke_survey



Scan the code above or visit the link for more information.

What kind of study design is this?

Retrospective, cross-sectional study



International Journal of
*Environmental Research
and Public Health*



Article

Perceptions of and Responses to Wildfire Smoke Among New York State Residents: A Cross-Sectional Study

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Abstract: Exposure to wildfire smoke (WFS) is associated with detrimental physical and mental health. Periods of sustained WFS are predicted to increase with climate change, affecting populations globally. Using a retrospective cross-sectional study, we assessed perceptions of and responses to WFS in a cohort of New York State (NYS) residents in Summer 2023. Data were collected using an online survey from October to November 2023. Descriptive statistics summarized respondent experiences, while exploratory analyses identified high-risk populations using chi-square and *t*-tests. Our sample consisted of 609 primarily healthy, white, and well-educated individuals who spent most of their time in NYS during Summer 2023. Of the 99% that reported experiencing WFS, 92% received

Barrera, E. I., Hayden, A., Meredith, G., & Noel, C. A. (2025). *International Journal of Environmental Research and Public Health*, 22(2), 277. <https://doi.org/10.3390/ijerph22020277>

Our research question today

What is the association between
exposure to wildfire smoke, and
health symptom(s) experienced (*outcome*)
in New York State residents during summer 2023?

(with some nuance based on the data we collected)

Your turn: Brainstorm/outline a questionnaire

10 min: Work with a partner to draft questions that you could use to answer our research question → think about the type of data (categoric (nominal, ordinal) or numeric) each question will produce

You want to record and quantify:

- Exposure to wildfire smoke
- Health symptoms experienced during times of wildfire smoke
- Respondent characteristics including age, gender, race/ethnicity, pre-existing medical conditions, and home residence

... during summer 2023 using a retrospective, cross-sectional study design

In our study, we adapted an existing questionnaire

www.nature.com/scientificdata

SCIENTIFIC DATA

OPEN

A dataset on human perception of and response to wildfire smoke

DATA DESCRIPTOR

Mariah Fowler¹, Arash Modaresi Rad¹, Stephen Utych², Andrew Adams¹, Sanazsadat Alamian³, Jennifer Pierce⁴, Philip Dennison⁵, John T. Abatzoglou⁶, Amir AghaKouchak⁷, Luke Montrose⁸ & Mojtaba Sadegh^{1*}

Wildfire smoke presents a growing threat in the Western U.S.; economic systems in growing western communities suffer due to fires. While modelling wildfire activity and associated wildfire impacts has improved, understanding how people perceive and respond to wildfire smoke has received little attention. Understanding and incorporating human perception of wildfire smoke is critical, as decision-makers need such information to mitigate wildfire impacts. We randomly selected people (in-person) across the Boise Metropolitan area and University affiliates (online), collecting information about their perception of wildfire smoke as a hazard, and smoke-related health issues. This dataset provides a novel perspective of people's perception of wildfire smoke and policy-relevant information to decision-makers. Dataset is available to address a wide range of research questions.

Category	Questions content
Demographic Data (6 questions)	• Age, • Gender, • Race, • Zip code, • Education level, • Income
Activity Data (3 questions)	• General health status, • Engagement in outside activities, • Frequency of outside activities
Air Quality Notification (13 questions)	• Receiving/Seeking air quality information and its source, • Frequency of seeking air quality information, • Reducing outside activities, • Longest period of consecutive days to reduce outside activities, • Minimum air quality index that convinced to reduce/eliminate outside activities, • Effective warning content and delivery method, • Timing of warning, • Future mitigation planning
Natural Hazard Questions (3 questions)	• Perception of smoke as a hazard, • Comparison with other hazards such as tornadoes and hurricanes, • Evacuating home to prevent smoke impacts
Health Questions (3 questions)	• Smoke-related health experience, • Type of observed symptoms, • Mitigation strategies to reduce health issues

Fowler, M., Modaresi Rad, A., Utych, S. *et al.* A dataset on human perception of and response to wildfire smoke. *Sci Data* 6, 229 (2019). <https://doi.org/10.1038/s41597-019-0251-y>

Table 1. Summary of survey questions for 614 in-person and 1,623 online participants.

Our questions on exposure and outcome

During the summer of 2023, where did you spend a majority of your time?

Select a state

During the summer of 2023, did you experience one or more days that were smoky, or days where you felt the air quality was poor due to wildfire smoke?

Yes

No

Not sure

During the summer of 2023, think of the longest period of consecutive days where it was smoky where you were. How many consecutive days was it smoky?

0 days

1 day

2 days

3 days

4 days

5 days

6 days

7 days or more

Not sure

Did you have any of the following symptoms during or a few days after one of the smoke events in summer 2023? *[Click on all that apply]*

Wheezing or whistling in the chest

Itchy irritated or watery eyes

Sneezing or a runny nose

Desktop survey preview

Dry nose / sinus

A sore or irritated throat

A cold

A dry cough at night

A dry cough first thing in the morning

A dry cough at other times of the day

A wet cough (congestion in the chest or phlegm production)

Bronchitis

An asthma attack

Headache

Fatigue

Anxiety

Other (please specify)

Next up:

- Implementation tips for online questionnaires
- Developing analysis plans
- Data dictionaries
- Cleaning data + inclusion/exclusion criteria

Lessons learned (by me) when implementing online questionnaires open to the public

- Online platform: Qualtrics
 - “To maintain data integrity, we activated the following security measures in Qualtrics: within question verification of geographic location, bot detection, security scan monitors, relevant ID, and indexing prevention.”
 - Can see completion time
- Targeted recruitment: emails from trusted leaders (NYS public health agencies, Cornell University, and Cornell Cooperative Extension) and via social media
- Raffle for compensation (\$50 Amazon gift cards)



Where did you live during summer 2023?

City/ Town

State

Zip Code

County

Best practice: Develop analysis plan *before* you analyze data, based on research/practice goals of project

Goal:

1. Estimate prevalence of health symptoms during times of WFS in summer 2023 in NYS population
 2. Estimate association between consecutive days of WFS and health symptoms experienced in sample
- Apply inclusion/exclusion criteria
 - Descriptive (or, univariate)
 - Distributions of key variables
 - Summary statistics: N and % (categorical); mean/median and SD + range (numeric)
 - Bivariate
 - T-tests (1 categorical, 1 numeric) and Chi-square/Fishers exact (2 categorical)
 - Multivariable model
 - Anticipated binary distribution (yes/no for each symptom) with non-rare outcomes
 - Log-binomial model, with log-link function models the probability of the outcome → exponentiating coefficients = prevalence ratios
 - Assumptions: independent observations + no interaction between exposure and covariates
 - Test/control for confounding: Add each potential confounder to the primary model and assess % change in the adjusted prevalence ratio from the crude estimate; include confounders with $\geq 10\%$ change in final model
 - Test for/include independent predictors of the outcome ($p < 0.05$)

Best practice: Create a data dictionary

What is a data dictionary?

- A structured document that defines each variable in a dataset
- Serves as a reference for data analysis + interpretation

What is typically included?

- Variable name (as stored in the dataset)
- Variable label / description (what the variable represents)
- Data type (e.g., categorical, numeric, ..)
- Values / response options (including coding)

...

Variable name from Qualtrics	Renamed as (in processing)	Description	Data type	Values / coding
“During the summer of 2023, where did you spend the majority of your time?”		State where respondent spent the majority of summer 2023	Categorical (nominal)	US state names from drop-down list (Alabama, Alaska, Arizona, ...)

Variable name from Qualtrics	Renamed as (in processing)	Description	Type	Values / Coding
“During the summer of 2023, where did you spend the majority of your time?”		State where respondent spent the majority of summer 2023	Categorical (nominal)	US state names from drop-down list (Alabama, Alaska, Arizona, ...)
“During the summer of 2023, did you experience one or more days that were smoky, or days where you felt that the air quality was poor due to wildfire smoke?”		Experienced one or more smoky days or days with poor air quality due to wildfire smoke during summer 2023	Categorical (nominal)	Yes; No; Not sure
“During the summer of 2023, think of the longest period of consecutive days where it was smoky where you were. How many consecutive days was it smoky?”		Longest period of consecutive smoky days during summer 2023	Categorical (ordinal)	0 days; 1 day; 2 days; 3 days; 4 days; 5 days; 6 days; 7 days or more; Not sure
“Did you have any of the following symptoms during or a few days after one of the smoke events in summer 2023? Check all that apply. Selected choice.”		Symptoms experienced during or a few days after a wildfire smoke event	Categorical (nominal; multiple response; stored as list)	Wheezing or whistling in the chest; Itchy, irritated, or watery eyes; Sneezing or runny nose; Dry nose or sinus irritation; Sore or irritated throat; A cold; Dry cough at night; Dry cough first thing in the morning; Dry cough at other times of the day; Wet cough (congestion in the chest or phlegm production); Bronchitis; An asthma attack; Headache; Fatigue; Anxiety; Other

After data collection is complete: Data cleaning

1. Clean + rename variables (if needed)

- `janitor: clean_names()`
 - “During the summer of 2023, think of the longest period of consecutive days where it was smoky where you were. How many consecutive days was it smoky?” →
`during_the_summer_of_2023_think_of_the_longest_period_of_consecutive_days_where_it_was_smoky_where_you_were_how_many_consecutive_days_was_it_smoky`
- `dplyr: rename(days_longest_conseq_wfs =
during_the_summer_of_2023_think_of_the_longest_period_of_consecutive_days_where_it_was_smoky_where_you_were_how_many_consecutive_days_was_it_smoky)`

2. Remove variables not needed for analysis (optional)

- `dplyr: select()`

3. Processes variables into forms needed for analysis (if needed)

- `symptom_list` → individual binary (yes/no) variables for each symptom on checklist
- `dplyr: mutate(chest_wheeze = case_when(
str_detect(symptom_list, "Wheezing or whistling in the chest") ~ "Yes",
TRUE ~ "No"))`

4. Explore to find any data input or recording errors → Descriptive statistics

5. Limit dataset to final analytical sample → Inclusion + exclusion criteria

6. Manipulate dataset format (if needed)

- `dplyr: pivot_wider()` and `pivot_longer()`
`long_data_symptoms = filtered_data %>%
select(id, symptom_any:anxiety) %>%
mutate(id = as.factor(id)) %>%
pivot_longer(-id, names_to = "symptoms", values_to = "count") %>%
mutate(symptoms = as.factor(symptoms)) %>%
mutate(count = fct_rev(count))`

Inclusion / exclusion criteria

“Through recruitment, a total of 1246 survey responses were collected. However, responses were excluded from final analysis if they met any of the following exclusion criteria:

1. Respondent listed a residence outside of NYS during Summer 2023;
2. Qualtrics system flagged a response as a bot;
3. Qualtrics system flagged a response as a duplicate entry;
4. Fewer than 50% of the survey questions were answered;
5. Respondent spent less than 120s on the survey;
6. The state, county, and city provided did not match any location within NYS; or
7. State and county or zip code with city was not provided”

Your turn: Practice applying exclusion criteria

First: Load dataset:

“WFS_survey_download_VTPEH6105.csv” + review data dictionary

- *Note: this dataset is partially processed (by me). I already renamed variables (see data dictionary), limited the dataset to variables needed for analysis, and created individual binary (Y/N) variables for each symptom checked.*

Your tasks → write R code to:

1. Explore respondents who meet each exclusion criteria below:
 - a) Identify number of respondents who Qualtrics flagged as a bot, or as a duplicate entry;
 - b) Identify number of respondents who spent <120 seconds on the survey, or answered <50% of questions;
 - c) Identify number of respondents who listed a residence outside of NYS during summer 2023
2. Apply exclusion criteria to limit to final analysis sample.
 - How many respondents are included in your sample, after excluding respondents using the above criteria?

ANSWERS: Practice applying exclusion criteria

Read in data: `wfs_data_not_filtered = read_csv("Data/WFS_survey_download_VTPEH6105.csv")`

Write R code to:

1. Explore respondents who meet each exclusion criteria:
 - a) Identify number of respondents who Qualtrics flagged as a bot (**n=147**), or as a duplicate entry (**n=149**) → combined criteria: **n=288**
 - b) Identify number of respondents who spent <120 seconds on the survey (**n=36**), or answered <50% of questions (**n=46**) → combined criteria: **n=56**
 - c) Identify number of respondents who listed a residence outside of NYS (**n=357 using state_23 or n=409 using state**) → combined criteria: **n=452**
2. Apply exclusion criteria to limit to final analysis sample. How many respondents are included in your sample, after excluding respondents using the above criteria?
 - **Respondents in sample, after above exclusion criteria: n= 691**
 - `wfs_data_filtered <- wfs_data_not_filtered %>%
 filter(
 q_recaptcha_score >= 0.5,
 is.na(q_relevant_id_duplicate),
 duration_in_seconds >= 120,
 progress >= 50,
 state_su23 == "NY - New York" | str_to_upper(state) %in% c("NEW YORK", "NY"))`

Inclusion / exclusion criteria

“Through recruitment, a total of 1246 survey responses were collected. However, responses were excluded from final analysis if they met any of the following exclusion criteria:

1. Respondent listed a residence outside of NYS during Summer 2023;
2. Qualtrics system flagged a response as a bot;
3. Qualtrics system flagged a response as a duplicate entry;
4. Fewer than 50% of the survey questions were answered;
5. Respondent spent less than 120s on the survey;
6. The state, county, and city provided did not match any location within NYS; or
7. State and county or zip code with city was not provided

After applying all exclusion criteria and cross-checking geographical locations using zip codes and municipality names, our sample consisted of 609 responses.”

→ For this analysis, we exclude also any respondents who are missing data for any of our variables, resulting in a final analytical sample of 555 responses.

Next up: Implementing our analysis plan

- Descriptive
- Bi-variate
- Multivariable, with assessment of confounding
- Main findings + interpretation

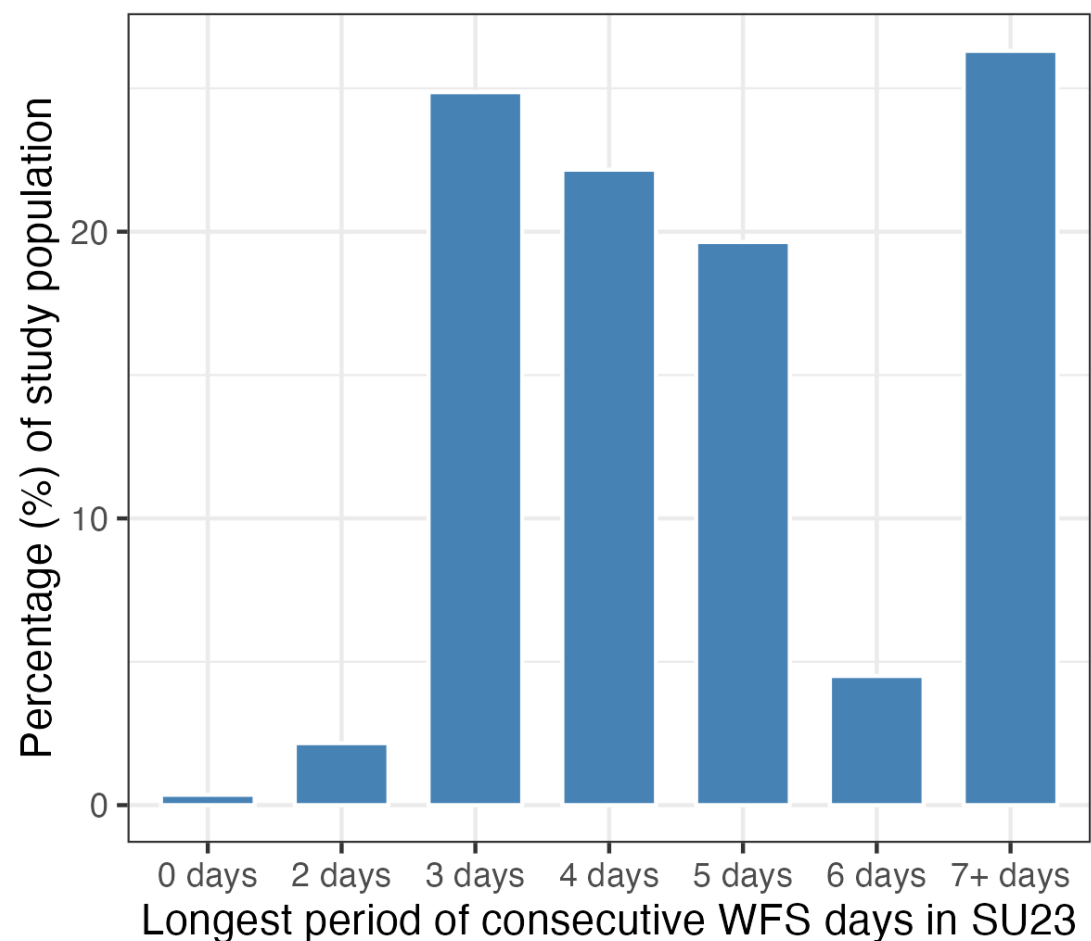
What is the goal of descriptive analysis?

- Understand data + study population
 - Any errors or unexpected entries in data?
 - Do our initial analyses make sense?
- Set basis for more advanced modeling

For our study:

- Distribution/visualization of key variables
 - days_longest_conseq_wfs (*exposure*)
 - each symptom (*outcome*)
 - age, gender, race_ethn, education, and general_health (*potential confounders*)
- Summary statistics for key variables: N and % (categorical); mean/median and SD + range (numeric)
- Describe study population – table or visual(s)

Descriptive stats: Example of categoric (days_longest_conseq_wfs)



days_longest_conseq_wfs	n	percent
0 days	2	0.40%
2 days	12	2.20%
3 days	138	24.90%
4 days	123	22.20%
5 days	109	19.60%
6 days	25	4.50%
7+ days	146	26.30%

```
wfs_data_for_analysis = read_csv("Data/WFS_survey_data_for_analysis_VTPEH6105.csv")

wfs_data_for_analysis %>%
  count(days_longest_conseq_wfs) %>%
  mutate(days_longest_conseq_wfs = recode(days_longest_conseq_wfs,
                                          "7 days or more" = "7+ days"),

  pct_numeric = n / sum(n) * 100,
  percent = paste0(round(pct_numeric, 1), "%")) %>%
  ggplot(aes(x = factor(days_longest_conseq_wfs), y = pct_numeric)) +
  geom_bar(stat = "identity", fill = "steelblue", color = "white", width = 0.7) +
  labs(
    x = "Longest period of consecutive WFS days in SU23",
    y = "Percentage (%) of study population") +
  theme_bw()
ggsave("Figures/WFS_days_distr.png", height=3.5, width=4)

wfs_data_for_analysis %>%
  tabyl(days_longest_conseq_wfs, ) %>%
  adorn_pct_formatting(1) %>%
  write_csv("Tables/WFS_consec_days_freq.csv")
```

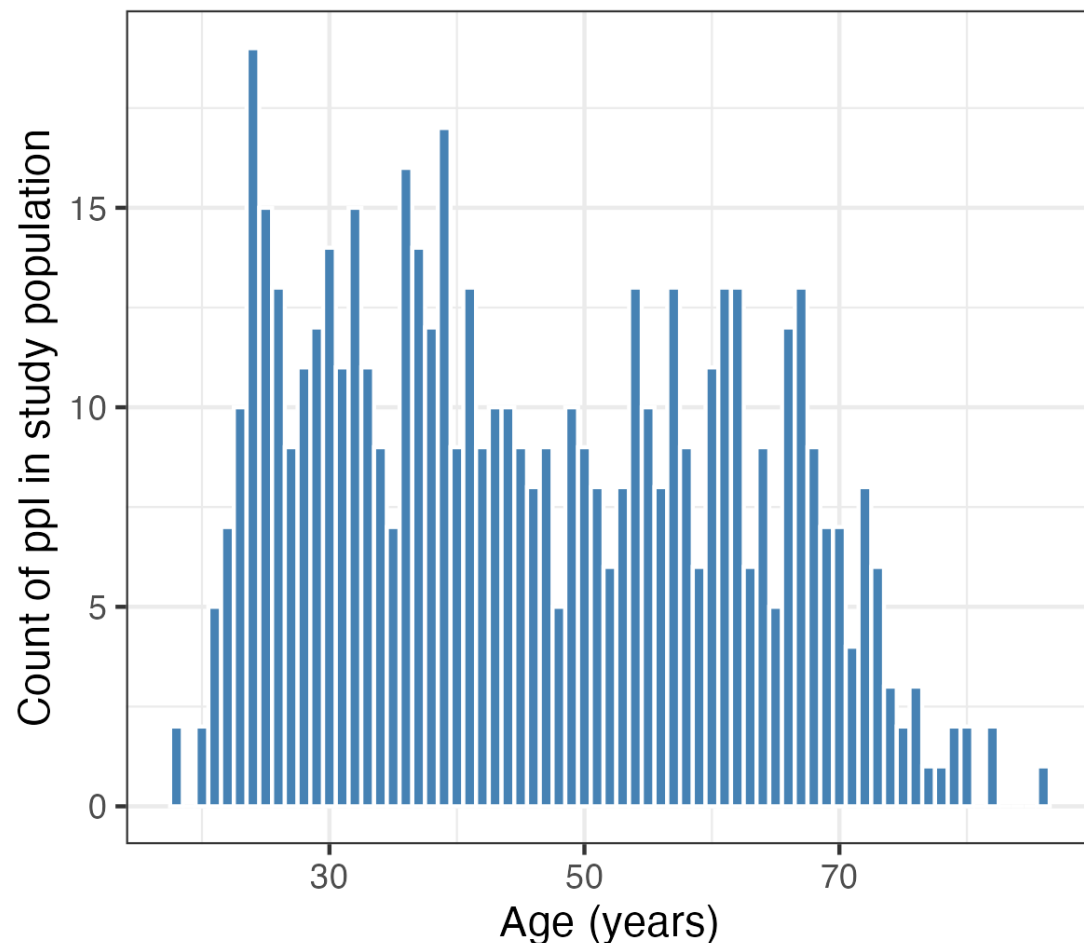
My thoughts as I look at these data:

- What happened to "1 day" WFS response?
- What is most common response?
- Ordinal variable... should I analyze as numeric or categoric?
- If numeric, does it look like a certain distribution?
- If categoric, should I collapse the categories?

Descriptive stats: Example of numeric (age)

Age = numeric (but was entered as free-text)

```
> # Free text entry on questionnaire - what was entered?
> wfs_data_for_analysis %>%
+   count(age, sort = TRUE) %>%
+   print(n = Inf)
# A tibble: 66 x 2
```



n	n_missing	mean	sd	min	max
553	2	45.9	15.9	18	86

```
wfs_data_for_analysis %>%
  mutate(age = as.numeric(age)) %>% #will coerce any non-numeric into NA
  ggplot(aes(x = age)) +
  geom_histogram(fill = "steelblue", color = "white", binwidth = 1) +
  labs(
    x = "Age (years)",
    y = "Count of ppl in study population") +
  theme_bw()
ggsave("Figures/age_distr.png", height = 3.5, width = 4)

wfs_data_for_analysis %>%
  mutate(age = as.numeric(age)) %>% #will coerce any non-numeric into NA
  summarise(
    n      = sum(!is.na(age)),
    n_missing = sum(is.na(age)),
    mean    = round(mean(age, na.rm = TRUE), 1),
    sd      = round(sd(age, na.rm = TRUE), 1),
    min     = min(age, na.rm = TRUE),
    max     = max(age, na.rm = TRUE)
  ) %>%
  write_csv("Tables/age_summary_stat.csv")
```

My thoughts as I look at these data:

- Was anything non-numeric entered? If so, what should I do?
- Is the min and max what I expect? Any outliers?
- What distribution does it resemble?
- Should I treat as numeric or collapse into categories?

... Completed for all key predictors

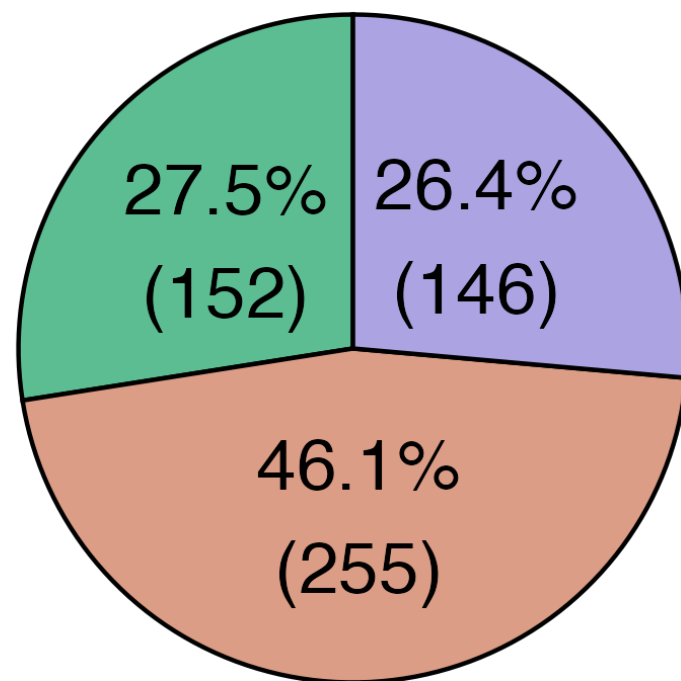
Through descriptive analyses, made the following decisions:

1. Collapse select categorical variables
 - Consecutive days of WFS → *wfs_wave_cat* (≤ 3 days, 4 to 6 days, 7+days)
 - General health → *excellent_health* (yes, no)
 - Education → *edu_BS_or_more* (yes, no)
 - Race/ethnicity → *cauc* (yes, no)
2. Delete entries where numeric age not specified
3. Can treat age as a normally distributed numeric/continuous variable

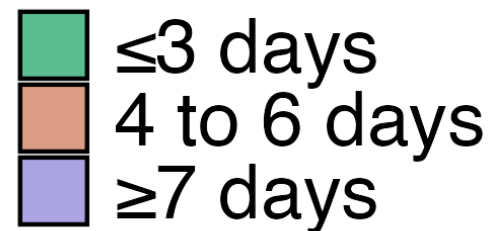
Description of our study population

Convenience sample: **553 New York State residents**

- Diverse ages ranging from 18 to 86 (mean: 45.9)
- Primarily Caucasian (88.8%) and identify as woman (82.6%)
- Most reported excellent (29.5%) or good health (58.9%)
- Most have college degree or higher (74.7%)
- 99.3% experienced any WFS in summer 2023



Longest period of consecutive wildfire smoke days



Your turn

	Symptom
	symptom_any
1	chest_wheeze
2	eyes_itchy_watery
3	nose_sneeze_runny
4	nose_sinus_dry
5	throat_sore
6	cold
7	cough_dry_night
8	cough_dry_morning
9	cough_dry_other
10	cough_wet
11	bronchitis
12	asthma_attack
13	headache
14	fatigue
15	anxiety

Each row assigned a health symptom (our outcomes)

First: Load + review analysis dataset

“WFS_survey_data_for_analysis_VTPEH6105.csv”

- *Note that this has been pre-filtered using our exclusion criteria, with the 2 entries where age was not numeric deleted (total n=553); Collapsed categorical variables created: wfs_wave_cat, excellent_health, cauc, edu_BS_or_more*

Your tasks → Write R code to:

1. Examine the distribution/visualize your assigned symptom
2. Calculate summary statistics, as appropriate
 - N and % (categorical)
 - mean/median and SD + min/max (numeric)
3. Report out: Enter summary statistics into Google spreadsheet + think about interpretation

Link to spreadsheet: bit.ly/41CX8yj



ANSWERS: Prevalence of symptoms, ranked by frequency

symptoms	Yes	No
symptom_any	88.6% (490)	11.4% (63)
eyes_itchy_watery	62.4% (345)	37.6% (208)
throat_sore	49.9% (276)	50.1% (277)
headache	48.6% (269)	51.4% (284)
nose_sneeze_runny	33.5% (185)	66.5% (368)
nose_sinus_dry	26.0% (144)	74.0% (409)
chest_wheeze	21.0% (116)	79.0% (437)
fatigue	21.0% (116)	79.0% (437)
anxiety	18.8% (104)	81.2% (449)
cough_dry_other	16.1% (89)	83.9% (464)
cough_dry_night	14.8% (82)	85.2% (471)
cough_dry_morning	11.6% (64)	88.4% (489)
asthma_attack	6.3% (35)	93.7% (518)
cough_wet	3.3% (18)	96.7% (535)
cold	2.0% (11)	98.0% (542)
bronchitis	1.4% (8)	98.6% (545)

What is the goal for bivariate analyses?

- Understand your data
 - Relationship between key variables, including multicollinearity
 - Sometimes included in Table 1
- Informs advanced model building
- Sometimes it is the end goal – although can be misleading

For our study:

- Relationship between key variables
 - Outcome(s): each symptom
 - Exposure: *wfs_wave_cat*
 - Potential confounders: *age*, *gender*, race/ethnicity (*cauc*), education (*edu_BS_or_more*), and general_health (*excellent_health*)
- T-tests or ANOVA (1 categoric, 1 numeric) and Chi-square or Fishers exact (2 categoric)

Bivariate analyses: Example examining the association between 2 categoric variables

wfs_wave_cat	Any symptom reported	
	Yes	No
3 or less days	124 (81.6%)	28 (18.4%)
4 to 6 days	227 (89.0%)	28 (11.0%)
7+ days	139 (95.2%)	7 (4.8%)

X-squared = 13.778
p-value = 0.001019

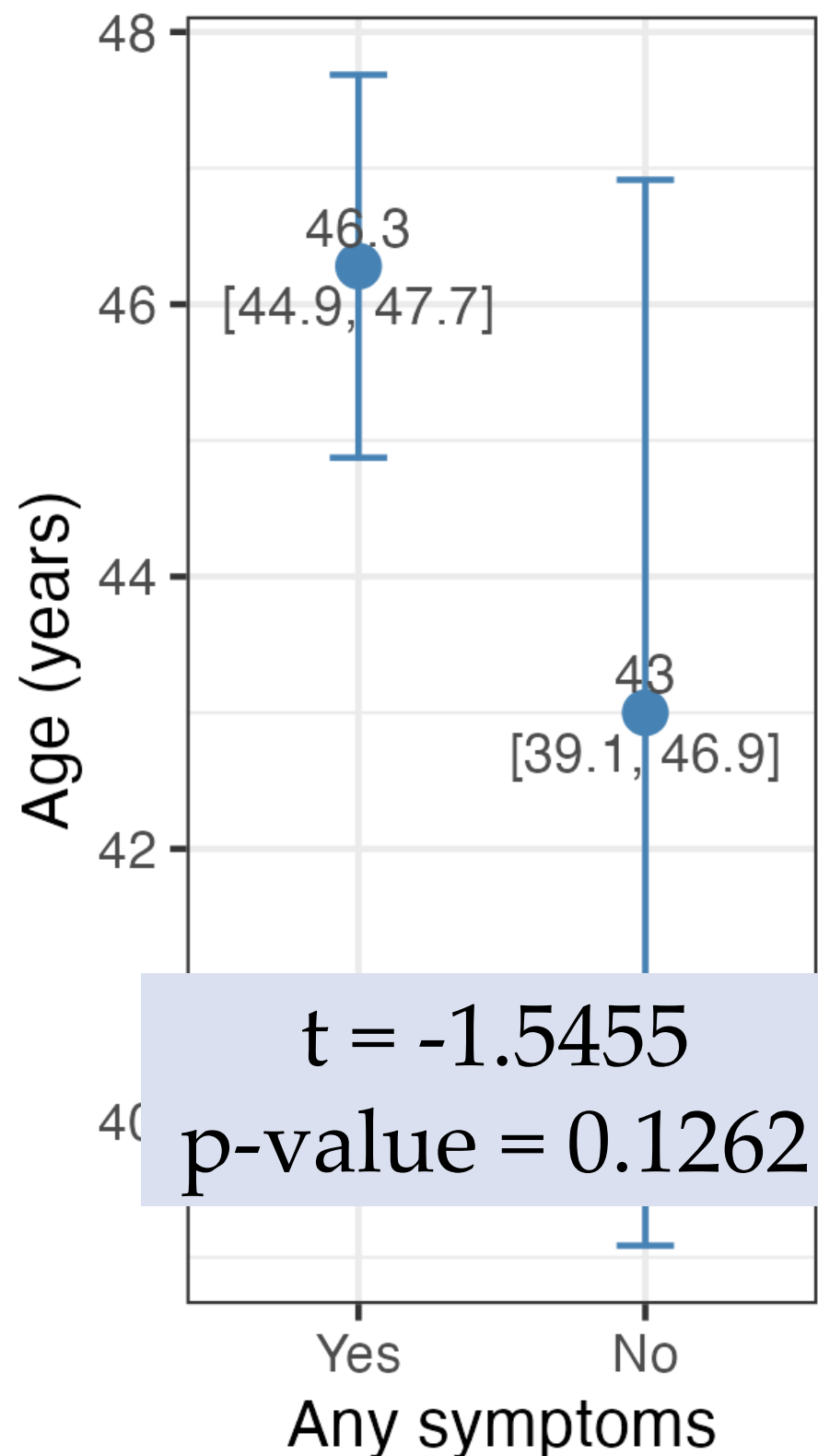
```
#Table for export
any_sym_wfs_table <- wfs_data_final %>%
  tabyl(wfs_wave_cat, symptom_any) %>%
  adorn_percentages("row") %>%
  adorn_pct_formatting(1) %>%
  adorn_ns(position = "front") %>%      # counts first, % in par
  select(wfs_wave_cat, Yes, No) %>%    # Yes as first column
  write_csv("Tables/WFS_wave_any_symp.csv")
any_sym_wfs_table

# Chi-square test
wfs_data_final %>%
  tabyl(wfs_wave_cat, symptom_any) %>%
  chisq.test() #X-squared = 13.778, df = 2, p-value = 0.001019
```

My thoughts as I look at these data:

- It looks like *wfs_wave_cat* is significantly (at $p < 0.05$) associated with reporting any symptom.. Will this association hold in advanced models?
- Appears that those reporting more consecutive WFS days report a higher prevalence of health symptoms but need to do post-hoc analyses to confirm.

Bivariate analyses: Example examining the association between 1 categoric and 1 numeric variable



Plot shows means +/- 95% CI

```
#visual:
wfs_data_final %>%
  mutate(symptom_any = fct_relevel(symptom_any, "Yes", "No")) %>%
  group_by(symptom_any) %>%
  summarise(
    mean = mean(age, na.rm = TRUE),
    se    = sd(age, na.rm = TRUE) / sqrt(n()),
    ci_lower = mean - 1.96 * se,
    ci_upper = mean + 1.96 * se
  ) %>%
  ggplot(aes(x = symptom_any, y = mean)) +
  geom_point(size = 3, color = "steelblue") +
  geom_errorbar(aes(ymin = ci_lower, ymax = ci_upper), width = 0.2, color = "steelblue") +
  geom_text(aes(label = paste0(round(mean, 1),
                                "\n[", round(ci_lower, 1), ", ", round(ci_upper, 1), "]")),
            hjust = 0.5, size = 3.2, color = "gray30") +
  labs(x = "Any symptoms", y = "Age (years)") +
  theme_bw()
ggsave("Figures/age_any_symptom_means.png", height = 3.5, width = 2)

#Welch's t-test (default in R) -- does not assume equal variances, handles unequal sample sizes
t.test(wfs_data_final$age ~ wfs_data_final$symptom_any)
```

My thoughts as I look at these data:

- There may be a relationship between *age* and reporting any symptoms, but not significant at $p < 0.05$ threshold... lots of variation in age in those reporting "no".
- Age might be influential in advanced models, but we will have to see (with outlined analysis plan).

Your turn

For your assigned symptom, write R code to examine the bivariate relationship between symptom (outcome) and:

- a) wfs_wave_cat* (3-level variable for consecutive days of WFS: ≤ 3 days, 4 to 6 days, 7+days)
- b) age* (in years, numeric)
- c) excellent_health* (yes, no)

Questions to think about for each bivariate analysis:

- What is the correct test to run for the type of variables you are examining?
- What is the magnitude and direction of the association, and p-value?
- What might this mean for our more advanced multivariable model?

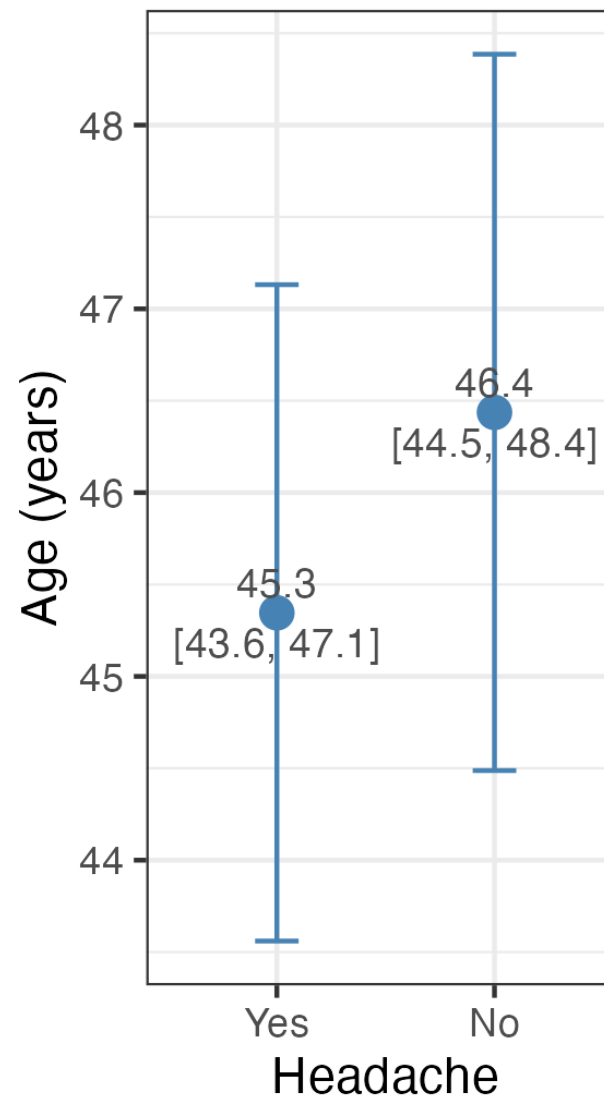
ANSWERS: headache as example

WFS

wfs_ wave_ cat	Headache reported	
	Yes	No
< 3 days	52 (34.2%)	100 (65.8%)
4 to 6 days	126 (49.4%)	129 (50.6%)
7+ days	91 (62.3%)	55 (37.7%)

X-square = 23.68
p-value < 0.001

Age



t = 0.809
p-value = 0.419

Health

excellent_ health	Headache reported	
	Yes	No
Yes	65 (39.9%)	98 (60.1%)
No	204 (52.3%)	186 (47.7%)

X-square = 6.621
p-value = 0.010

Finally, multivariable analysis

- What does multivariable analysis allow us to do?
 - Isolate individual effects of each predictor
 - Control for confounding

For our study:

- Use log-binomial model to estimate association between exposure to WFS (consecutive days) and self-reported symptoms in times of WFS → prevalence ratios + 95% CI
 - Primary (crude) analysis: symptom ~ WFS_wave_cat
 - Test for confounding:
 - Does potential confounder associate with exposure? (bivariate)
 - Does potential confounder associate with outcome? (bivariate)
 - Does crude effect estimate change by $\geq 10\%$ when adjusting for potential confounder?
 - Test for independent predictors of the outcome
 - Final model: symptom ~ WFS_wave_cat + (any confounders)

Example: Headache as symptom

Crude model:

```
# Log-binomial model: headache ~ WFS wave category  
fit_model <- logbin(as.numeric(headache == "Yes") ~ wfs_wave_cat,  
                    data = wfs_data_final,  
                    method = "em")
```

		PR	95% CI	
Consecutive days of WFS	3 or less days		<i>ref</i>	
	4 to 6 days	1.444	1.121	1.86
	7+ days	1.822	1.413	2.349

Age, gender, race/ethnicity, education, health status

Include other variables in model if:

- (1) Adjustment changes effect estimate(s) by >10%, or
- (2) Statistically significant predictor of the outcome

Example: Headache as symptom

Testing for confounding and independent predictors of the outcome

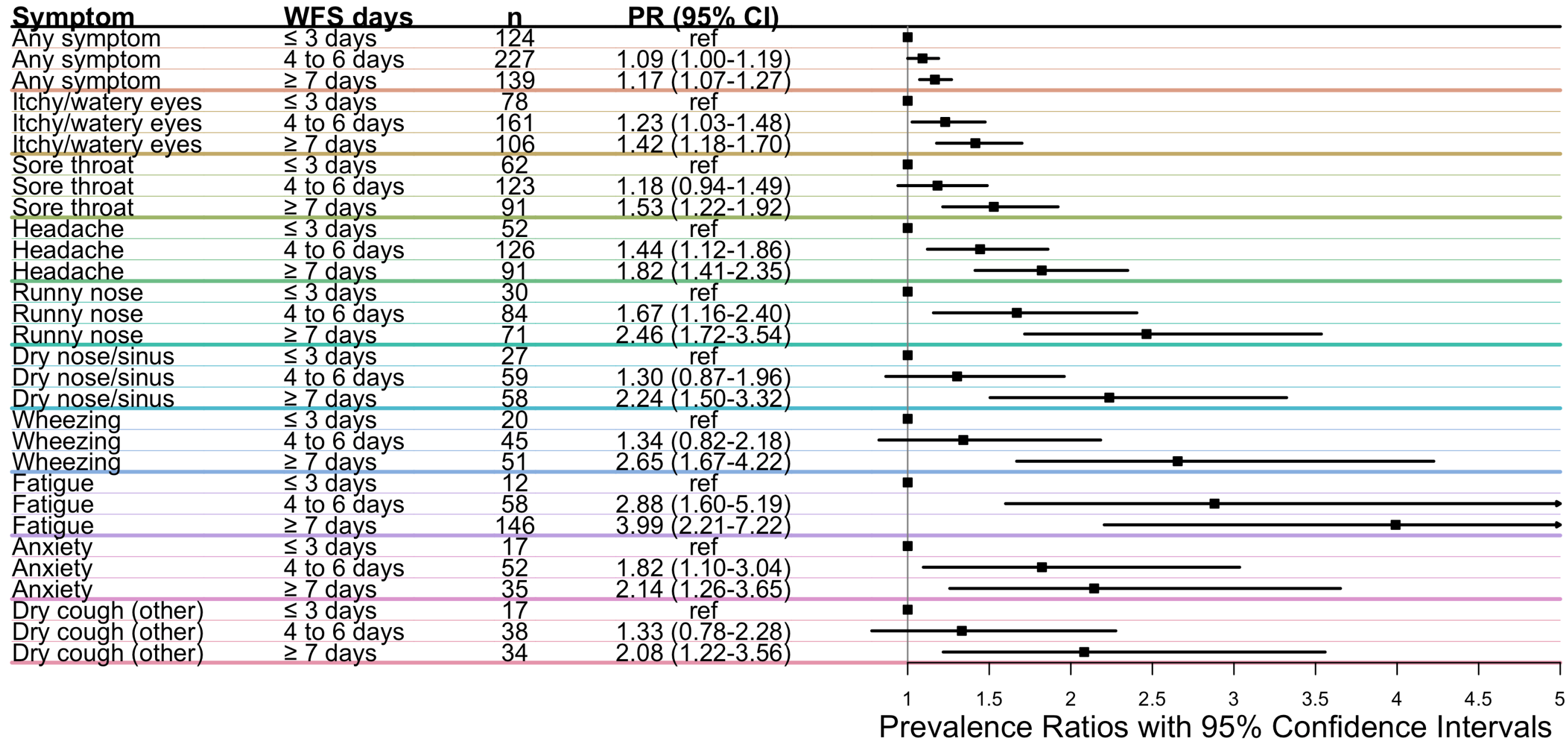
Example: Headache as symptom

Final model:

```
fit_model <- logbin(as.numeric(headache == "Yes") ~ wfs_wave_cat #exposure of interest
+ gender #not confounder, but independent predictor of outcome
+ edu_BS_or_more #not confounder, but independent predictor of outcome
+ excellent_health, #not confounder, but independent predictor of outcome
data = wfs_data_final,
method = "em")
```

		PR	95% CI	
Consecutive days of WFS	3 or less days		ref	
	4 to 6 days	1.406	1.10	1.80
	7+ days	1.733	1.35	2.22
Health status	Not excellent		ref	
	Excellent	0.798	0.65	0.98
Gender	Man		ref	
	Woman	1.817	1.31	2.52
Education level	<B.S.		ref	
	BS+	0.820	0.70	0.96

What are the main findings?



Note: Gender, age, health status, race/ethnicity, and education level were considered as potential confounders. Only variables that changed the crude PR > 10% were included in the final models. (none did)

Discussion

Main findings

- 88.6% reported ≥ 1 health symptom during WFS
 - Top 3: itchy/watery eyes (62.4%), sore throat (49.9%), and headache (48.6%)
- Persistent trend of increased symptom prevalence with longer periods of wildfire smoke
 - Symptom prevalence 9.1% and 16.7% higher for 4 to 6 and ≥ 7 days, versus ≤ 3 consecutive WFS days
 - Control for potential confounders did not appreciably change estimates

Limitations

- Lack of objective measure of WFS exposure
- Potential for recall bias
- Sample is not representative of NYS population
- Cross-sectional data limits causal inference

Today we discussed...

1. Common types of questions on questionnaires + why this matters for analysis
2. Different types of variables (categorical vs. numeric)
3. The importance of developing analysis plans prior to analysis (ideally prior to collecting data)
4. Implementation tips for online questionnaires
5. The development of data dictionaries
6. Cleaning data + applying inclusion/exclusion criteria
7. Used data from a retrospective, cross-sectional WFS questionnaire: Descriptive, bivariate, + multivariable models with control for confounding



Any questions?

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